

Generalization near a bifurcation point

Science, Structure, Details, Tone – pedagogical, Figures – CCC, Formatting

1 Introduction

Understanding how generative models generalize near critical phenomena—such as bifurcations, phase transitions, and scale-invariant fluctuations—is a central challenge in machine learning for the physical sciences. In critical regimes, system dynamics are marked by diverging variance and persistent temporal memory (critical slowing down). This study evaluates the theoretical capacity of deep neural networks—specifically score-based generative models—to learn and generalize near critical points when trained exclusively using off-critical data.

2 Linear Toy Model

2.1 Data-generating process

Conditional distribution. For each r , samples $\mathbf{x} \in \mathbb{R}^d$ are drawn from

$$\mathbf{x} \mid r \sim \mathcal{N}(\mathbf{0}, \Sigma(r)). \quad (1)$$

Critical covariance. Let $\Sigma(r)$ be the covariance of the distribution given in (1).

Since the covariance has a symmetric positive semi-definite structure it has an eigen decomposition as follows,

$$\Sigma(r) = Q(r)\Lambda(r)Q(r)^T, \quad \Lambda(r) = \text{diag}(\lambda_1(r), \dots, \lambda_d(r)) \quad (2)$$

where the orthonormal columns of $Q(r) = [\mathbf{e}_1(r), \mathbf{e}_2(r), \dots, \mathbf{e}_d(r)]$ define the **principal fluctuation directions(eigenvectors)** in state space $\mathbb{R}^d$, and $\lambda_k(r)$ represents the variance along direction $\mathbf{e}_k(r)$.

Diverging critical mode. Near the bifurcation critical point r_c , the system undergoes critical fluctuations where variance along the primary Order Parameter mode—the critical direction $\mathbf{e}_1(r)$ —diverges, while fluctuations along all transverse modes $\mathbf{e}_{k \geq 2}$ remain bounded.

Mathematically, as $r \rightarrow r_c$:

$$\lambda_1(r) \sim |r - r_c|^{-\nu},$$

while the other eigenvalues remain bounded:

$$\lambda_k(r) = O(1), \quad k = 2, \dots, d.$$

Here, "direction" refers specifically to the principal eigenvector $\mathbf{e}_1$ corresponding to the dominant eigenvalue $\lambda_1(r)$. Along this eigenvector, variance grows parametrically faster than in any orthogonal subspace:

$$\frac{\lambda_k(r)}{\lambda_1(r)} \rightarrow 0 \quad (k \geq 2)$$

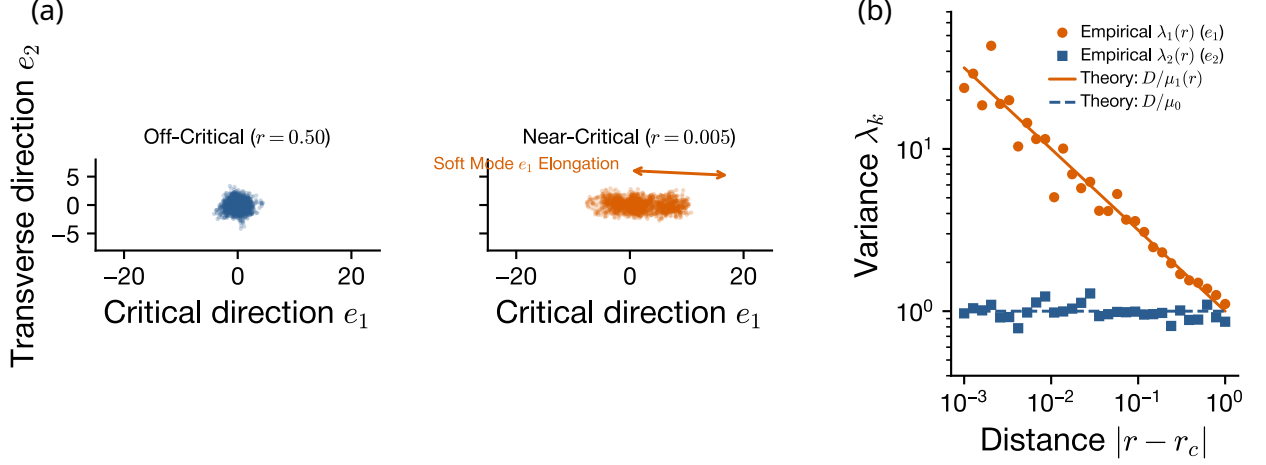


Figure 1: **(a) Dynamical origin of spatial anisotropy.** Steady-state sample distributions in state space generated by the multidimensional Langevin process $dx(t) = -\nabla_x V(x, r) dt + \sqrt{2D} dW(t)$ off-criticality ($r = 0.50$, left) and near-criticality ($r = 0.005$, right). As the control parameter $r \rightarrow r_c$, the restoring force along the critical mode vanishes ($\mu_1(r) \rightarrow 0$), driving anisotropic fluctuation growth and severe elongation along the soft mode direction e_1 , while fluctuations along the transverse direction e_2 remain tightly bounded. Shaded regions depict the 2σ spatial covariance ellipses. **(b) Empirical covariance spectrum.** Eigenvalues of the empirical covariance matrix Σ plotted against the distance to criticality $|r - r_c|$. The dominant eigenvalue $\lambda_1(r)$ diverges as $D/\mu_1(r) \sim |r - r_c|^{-\nu}$ (with $\nu = 0.50$), whereas transverse eigenvalues $\lambda_{k \geq 2}$ remain stationary ($D/\mu_0 = O(1)$), confirming the dynamical emergence of a low-dimensional critical manifold.

One direction exhibits parametrically faster growth of variance than the others.

Without loss of generality, we choose our coordinate basis such that the critical direction coincides with the first canonical basis vector $\mathbf{e}_1 = (1, 0, \dots, 0)^\top$. Therefore the covariance matrix simplifies to,

$$\Sigma(r) = I_d + \frac{1}{|r - r_c|^\nu} \mathbf{e}_1 \mathbf{e}_1^\top, \quad (3)$$

where $\nu > 0$ is the critical exponent. The eigenvalues of $\Sigma(r)$ are

$$\sigma_1(r) = 1 + |r - r_c|^{-\nu}, \quad \sigma_k(r) = 1 \quad (k = 2, \dots, d).$$

As $r \rightarrow r_c$, the variance along $\mathbf{e}_1$ diverges as $|r - r_c|^{-\nu}$, mimicking the diverging fluctuations near a critical point. The critical exponent $\nu > 0$ governs the power-law divergence of variance along the order parameter mode, playing an analogous role to critical susceptibility exponents (γ) in continuous phase transitions or soft-mode instabilities in dynamical systems.

2.2 True score function

The score of a Gaussian $\mathcal{N}(\mathbf{0}, \Sigma)$ is computed by direct differentiation of its log-density. The density is

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2} \mathbf{x}^\top \Sigma^{-1} \mathbf{x}\right),$$

so

$$\log p(\mathbf{x}) = -\frac{1}{2} \mathbf{x}^\top \Sigma^{-1} \mathbf{x} + \text{const}, \quad \nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\Sigma^{-1} \mathbf{x}.$$

For our problem,

$$s^*(\mathbf{x}, r) \equiv \nabla_{\mathbf{x}} \log p(\mathbf{x} | r) = -\Sigma(r)^{-1} \mathbf{x}. \quad (4)$$

This inverse can be calculated and we get

$$\Sigma(r)^{-1} = I_d + \left(\frac{1}{\sigma_1(r)} - 1 \right) \mathbf{e}_1 \mathbf{e}_1^\top. \quad (5)$$

Critical direction: $1/\sigma_1(r) = |r - r_c|^\nu / (1 + |r - r_c|^\nu) \rightarrow 0$ as $r \rightarrow r_c$, so the score component along $\mathbf{e}_1$ vanishes at the bifurcation. **Non-critical directions:** the score is simply $-x_k$ for $k \geq 2$.

3 Student model and loss function

3.1 Teacher-Student Framework

The Teacher-Student framework is a classic analytical paradigm used to study learning, generalization, and structural inference in neural networks. In this framework, data generation and learning are divided into two distinct components:

- **The Teacher Model:** A generative mechanism or target mapping $p(x|\theta^*)$ parameterized by ground-truth latent variables or weights θ^* . The teacher defines the underlying data-generating distribution and dictates the true conditional statistics or scores of the system.
- **The Student Model:** An approximating neural network or estimator $s_\theta(x)$ parameterized by trainable weights θ . The student receives finite or domain-restricted samples generated by the teacher and optimizes a task-specific loss function (such as mean squared error or score matching) to reconstruct the teacher’s internal structure or predictive output.

By assuming an explicit teacher model, the theoretical performance of the student can be evaluated against the true data-generating process rather than empirical test sets alone. This framework is crucial to analyze how structural constraints on training data, such as domain gaps, missing features and finite sample sizes, impact the student’s parameter convergence and overall performance. With this as motivation we adopt the following appropriate student model for our problem.

3.2 Linear student model

Assuming (1) as the implicit teacher, we adopt a linear-in- $\mathbf{x}$ student with general r -dependence:

$$s_\theta(\mathbf{x}, r) = -A(r)\mathbf{x}, \quad A(r) \in \mathbb{R}^{d \times d}. \quad (6)$$

We parameterize $A(r)$ as a degree- $(K-1)$ polynomial in r in some basis $\{\phi_k(r)\}_{k=0}^{K-1}$ (monomials, Chebyshev, Legendre — the choice does not affect the result). While any complete polynomial basis spans the same finite-dimensional function space $\mathbb{P}_{K-1}$ at the exact theoretical limit ($P \rightarrow \infty$), using orthogonal polynomials ensures well-conditioned normal matrices and rapid spectral convergence during optimization, avoiding the severe ill-conditioning associated with standard monomial bases. For concreteness:

$$A(r) = \sum_{k=0}^{K-1} A_k \phi_k(r), \quad \phi_k(r) = r^k.$$

The trainable parameters are the K matrices $\{A_k\}$, totaling Kd^2 scalar parameters.

The (population) score-matching loss is **add a small schematic**

$$\mathcal{L}(\theta) = \mathbb{E}_{r \sim \rho_\Delta} \left[\mathbb{E}_{\mathbf{x}|r} \left[\|s_\theta(\mathbf{x}, r) - s^*(\mathbf{x}, r)\|^2 \right] \right], \quad (7)$$

where the training distribution over r has the gap excised:

$$\rho_{\Delta}(r) = \frac{1}{Z_{\Delta}} \mathbf{1}[|r - r_c| > \Delta/2] \cdot \rho_0(r), \quad (8)$$

with ρ_0 a base distribution (we take uniform on $[r_c - L, r_c + L]$) and Z_{Δ} the normalization.

Let $B(r) \equiv A(r) - \Sigma(r)^{-1}$. Then

$$\|s_{\theta}(\mathbf{x}, r) - s^*(\mathbf{x}, r)\|^2 = \|B(r)\mathbf{x}\|^2 = \mathbf{x}^{\top} B(r)^{\top} B(r) \mathbf{x}.$$

For any deterministic symmetric matrix M , the standard identity for a centered Gaussian $\mathbf{x} \sim \mathcal{N}(0, \Sigma)$ is $\mathbb{E}[\mathbf{x}^{\top} M \mathbf{x}] = \text{Tr}(M \Sigma)$. Hence

$$\mathbb{E}_{\mathbf{x}|r}[\|s_{\theta} - s^*\|^2] = \text{Tr}(B(r)^{\top} B(r) \Sigma(r)) = \text{Tr}(B(r) \Sigma(r) B(r)^{\top}).$$

So the loss becomes

$$\mathcal{L}(\theta) = \int \rho_{\Delta}(r) \text{Tr}\left[(A(r) - \Sigma(r)^{-1}) \Sigma(r) (A(r) - \Sigma(r)^{-1})^{\top}\right] dr. \quad (9)$$

3.3 Decoupling the dimensions

We now exploit the diagonal structure of $\Sigma(r)$. In the canonical basis,

$$\Sigma(r) = \text{diag}(\sigma_1(r), 1, 1, \dots, 1). \quad (10)$$

Which means

$$\Sigma(r)^{-1} = \text{diag}\left(\frac{1}{\sigma(r)}, 1, 1, \dots, 1\right) \quad (11)$$

The cost (9) can be expanded in matrix entries to give

$$\begin{aligned} \mathcal{L}(\theta) &= \int \rho_{\Delta}(r) \text{Tr}\left[(A(r) - \Sigma(r)^{-1}) \Sigma(r) (A(r) - \Sigma(r)^{-1})^{\top}\right] dr \\ &= \sum_{i=1}^d \sum_{j=1}^d (A_{ij}(r) - \Sigma_{ij}^{-1}(r))^2 \Sigma_{jj}(r) \end{aligned} \quad (12)$$

Because $\Sigma(r)^{-1}$ has no off-diagonal entries ($\Sigma_{ij}^{-1} = 0$ for $i \neq j$), any off-diagonal parameter $A_{ij}(i \neq j)$ contributes a non-negative term $A_{ij}(r)^2 \sigma(r) \geq 0$ to the loss. Since A_{ij} does not interact with any target diagonal terms, setting all off-diagonal components strictly to zero ($A_{ij} \equiv 0$ for $i \neq j$) globally minimizes the loss without constraining the diagonal fit. Writing $A(r) = \text{diag}(a_1(r), a_2(r), \dots, a_d(r))$ with each $a_k(r)$ a degree- $(K-1)$ polynomial, the trace splits as

$$\text{Tr}[\dots] = \sum_{k=1}^d (a_k(r) - 1/\sigma_k(r))^2 \sigma_k(r).$$

This reduces the matrix trace to a sum of d independent scalar loss functions:

$$\mathcal{L}(\theta) = \sum_{k=1}^d \mathcal{L}_k(a_k) = \sum_{k=1}^d \int \rho_{\Delta}(r) \left(a_k(r) - \frac{1}{\sigma_k(r)}\right)^2 \sigma_k(r) dr \quad (13)$$

3.4 Behavior of non-critical modes

For all non-critical directions ($k \geq 2$), the eigenvalues are r -independent constants: $\sigma_k(r) = 1$ and $1/\sigma_k(r) = 1$. The loss for these modes reduce to an unweighted error given by,

$$\mathcal{L}_k(a_k) = \int \rho_\Delta(r) (a_k(r) - 1)^2 dr \quad (14)$$

Since the constant target function $g_k(r) \equiv 1 (k \geq 2)$ lies directly inside the polynomial basis $\phi_k(r)_{k=0}^{K-1}$, the student easily achieves zero loss by setting $a_k(r) \equiv 1$. Thus the non-critical modes are learned trivially and contribute zero error to generalization near the bifurcation.

3.5 Behavior along the critical direction

All non-trivial learning dynamics collapse entirely into the critical direction $\mathbf{e}_1$. Dropping the subscript ($a(r) \equiv a_1(r)$), the loss in (13) simplifies to,

$$\mathcal{L}(a) = \int \rho_\Delta(r) \left(a(r) - \frac{1}{\sigma(r)} \right)^2 \sigma(r) dr \quad (15)$$

Further defining the effective *target function* $g(r)$ and the *weighting function* $w(r)$

$$g(r) \equiv \frac{1}{\sigma(r)} = \frac{|r - r_c|^\nu}{1 + |r - r_c|^\nu} \quad w(r) \equiv \rho_\Delta(r) \sigma(r) \quad (16)$$

we arrive at a *1D weighted least-squares polynomial fit*:

$$\mathcal{L}_1(a) = \int w(r) (a(r) - g(r))^2 dr. \quad (17)$$

of g on the support of ρ_Δ (i.e., on $|r - r_c| > \Delta/2$). **Expand this section.** Not sure how exactly to expand Note that

- For $|r - r_c| \ll 1$: $g(r) \approx |r - r_c|^\nu$ — the cusp.
- For $|r - r_c| \rightarrow \infty$: $g(r) \rightarrow 1$ — saturates.
- At $r = r_c$: $g(r_c) = 0$.

4 Minimising loss

We henceforth set $r_c = 0$ by relabeling. Expand $a(r) = \sum_{k=0}^{K-1} \theta_k r^k$, with $\boldsymbol{\theta} = (\theta_0, \dots, \theta_{K-1})^\top$ and $\boldsymbol{\phi}(r) = (1, r, r^2, \dots, r^{K-1})^\top$, so

$$a(r) = \boldsymbol{\theta}^\top \boldsymbol{\phi}(r).$$

The loss becomes a quadratic form in $\boldsymbol{\theta}$:

$$\begin{aligned} \mathcal{L}_1 &= \int w(r) (\boldsymbol{\theta}^\top \boldsymbol{\phi}(r) - g(r))^2 dr \\ &= \boldsymbol{\theta}^\top \left[\int w(r) \boldsymbol{\phi}(r) \boldsymbol{\phi}(r)^\top dr \right] \boldsymbol{\theta} - 2\boldsymbol{\theta}^\top \left[\int w(r) g(r) \boldsymbol{\phi}(r) dr \right] + \int w(r) g(r)^2 dr. \end{aligned} \quad (18)$$

Define

$$M_{kl} \equiv \int w(r) r^k r^l dr = \int \rho_\Delta(r) \sigma(r) r^{k+l} dr, \quad (19)$$

$$b_k \equiv \int w(r)g(r) r^k dr. \quad (20)$$

Crucially, $\sigma(r)g(r) = 1$, so the bias vector simplifies:

$$b_k = \int \rho_\Delta(r) r^k dr. \quad (21)$$

That is, b_k is the k -th moment of the *unweighted* training distribution ρ_Δ .

Setting $\partial\mathcal{L}_1/\partial\boldsymbol{\theta} = 0$ gives the normal equations $M\boldsymbol{\theta} = \mathbf{b}$, with solution

$$\hat{\boldsymbol{\theta}} = M^{-1}\mathbf{b}. \quad (22)$$

5 Scaling analysis

Place $r_c = 0$. Define a matching scale r_* of order unity that separates inner and outer regions:

- **Inner region:** $\Delta/2 < |r| < r_*$, with $|r|^\nu \ll 1$, so $\sigma_1(r) \approx |r|^{-\nu}$ and $g(r) \approx |r|^\nu$.
- **Outer region:** $r_* < |r| < L$, with $|r|^\nu \gtrsim 1$, so $\sigma_1(r) \rightarrow 1$ and $g(r) \rightarrow 1$.

5.1 Inner-region contributions

Assume $k + l$ even (only nonvanishing cases). The inner region contribution to M_{kl} (eqn.19) is

$$\begin{aligned} M_{kl}^{\text{in}}(\Delta) &= \int_{\Delta/2 < |r| < r_*} \rho_0(r) \sigma_1(r) r^{k+l} dr \\ &\approx 2\rho_0 \int_{\Delta/2}^{r_*} \left[\frac{1+r^\nu}{r^\nu} \right] r^{k+l} dr \\ &= \frac{2\rho_0}{k+l-\nu+1} \left[r_*^{k+l-\nu+1} - (\Delta/2)^{k+l-\nu+1} \right] + \frac{2\rho_0}{1+k+l} \left[r_*^{k+l+1} - (\Delta/2)^{k+l+1} \right] \end{aligned} \quad (23)$$

For $\nu < 1$ (~~Feigenbaum case~~) and $k + l \geq 0$ even, the exponent $k + l - \nu + 1 > 0$, so the integral is finite as $\Delta \rightarrow 0$ and the Δ -dependence enters as

$$M_{kl}^{\text{in}}(\Delta) = M_{kl}^{\text{in}}(0) - \frac{2\rho_0}{k+l-\nu+1} (\Delta/2)^{k+l-\nu+1} - \frac{2\rho_0}{k+l+1} (\Delta/2)^{k+l+1} \quad (24)$$

Similarly for eqn.21 the inner contribution is,

$$\begin{aligned} b_k^{\text{in}}(\Delta) &= \int_{\Delta/2 < |r| < r_*} \rho_0(r) r^k dr \\ &\approx \frac{2\rho_0}{k+1} \left[r_*^{k+1} - (\Delta/2)^{k+1} \right]. \end{aligned} \quad (25)$$

5.2 Outer-region contributions

For the outer region, $\sigma_1(r) \rightarrow 1$, $g(r) \rightarrow 1$, $w(r) \rightarrow \rho_0$. So the outer region contributions for (19) and (21) are:

$$M_{kl}^{\text{out}} \approx \rho_0 \int_{r_* < |r| < L} r^{k+l} dr, \quad b_k^{\text{out}} \approx \rho_0 \int_{r_* < |r| < L} r^k dr. \quad (26)$$

Cite other equations and figures as much as possible. Both are Δ -independent at this level of approximation.

5.3 Consolidated scaling of M and $\mathbf{b}$

Taking into account how M and $\mathbf{b}$ scale in the inner and outer regions as derived in (24), (25) and (??), the scaling of M and $\mathbf{b}$ in leading and subleading order of Δ can be arrived as:

$$M(\Delta) = M^{(0)} + \Delta^{1-\nu} M^{(1)} + O(\Delta) \quad (27)$$

and

$$\mathbf{b}(\Delta) = \mathbf{b}^{(0)} + \Delta \mathbf{b}^{(1)} + O(\Delta^2) \quad (28)$$

where $M^{(0)}, M^{(1)}, \mathbf{b}^{(0)}, \mathbf{b}^{(1)}$ are Δ -independent. The corrections in M scale as $\Delta^{1-\nu}$ because the leading inner integrand $\sim r^{-\nu}$ near the gap edge integrates to give a power of the gap that is one greater (boundary contribution). The corrections in $\mathbf{b}$ scale as Δ because the integrand is regular ($\sim r^0$).

We can write down the leading and sub-leading order of M and $\mathbf{b}$ in matrix form for more clarity:

$$\begin{aligned} M - M^{(0)} &= \begin{pmatrix} M_{00} - M_{00}^{(0)} & M_{01} - M_{01}^{(0)} & \cdots & M_{0K-1} - M_{0K-1}^{(0)} \\ M_{10} & M_{11} & \cdots & M_{1K-1} \\ \vdots & \vdots & \vdots & \vdots \\ M_{K0} & \cdots & \cdots & M_{K-1K-1} \end{pmatrix} \\ &\sim \begin{pmatrix} O(\Delta^{1-\nu}) & O(\Delta) & \cdots & O(\Delta^{K-\nu}) \\ O(\Delta) & O(\Delta^{3-\nu}) & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ O(\Delta^{K+1-\nu}) & \cdots & \cdots & O(\Delta^{2K-1-\nu}) \end{pmatrix} \end{aligned} \quad (29)$$

and

$$\mathbf{b} - \mathbf{b}^{(0)} = \begin{pmatrix} b_0 - b_0^{(0)} \\ b_1 - b_1^{(0)} \\ \vdots \\ b_{K-1} - b_{K-1}^{(0)} \end{pmatrix} \sim \begin{pmatrix} O(\Delta) \\ O(\Delta^2) \\ \vdots \\ O(\Delta^K) \end{pmatrix} \quad (30)$$

5.4 Expansion of $\hat{\theta}(\Delta)$

Since

$$\hat{\theta}(\Delta) = M^{-1} \mathbf{b},$$

then using (27) and (28) we have,

$$\begin{aligned} \hat{\theta}(\Delta) &= M(\Delta)^{-1} \mathbf{b}(\Delta) \\ &= (M^{(0)} + \Delta^{1-\nu} M^{(1)} + \cdots)^{-1} (\mathbf{b}^{(0)} + \Delta \mathbf{b}^{(1)} + \cdots) \\ &= \left(M^{(0)} (I + M^{(0)-1} [\Delta^{1-\nu} M^{(1)} + O(\Delta)]) \right)^{-1} (\mathbf{b}^{(0)} + \Delta \mathbf{b}^{(1)} + \cdots) \\ &= M^{(0)-1} \mathbf{b}^{(0)} + \Delta^{1-\nu} \left[-M^{(0)-1} M^{(1)} M^{(0)-1} \mathbf{b}^{(0)} \right] + O(\Delta). \end{aligned} \quad (31)$$

So

$$\hat{\theta}(\Delta) = \hat{\theta}^{(0)} + c_1 \Delta^{1-\nu} + O(\Delta), \quad (32)$$

where $\hat{\theta}^{(0)}$ is the limiting value at $\Delta \rightarrow 0$, with $\hat{\theta}^{(0)} = M^{(0)-1} \mathbf{b}^{(0)}$.

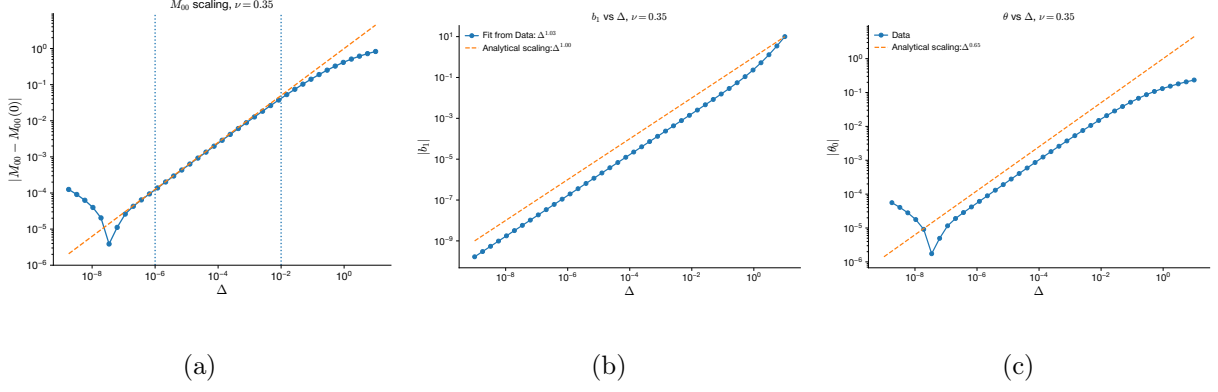


Figure 2: **Deterministic scaling properties of design matrix moments and critical parameters near the excised gap ($\nu = 0.35$).** (a) Absolute deviation of the leading moment matrix element $|M_{00}(\Delta) - M_{00}(0)|$ against the gap size Δ . The numerical trend verifies the theoretical inner-region scaling contribution of $\Delta^{1-\nu}$ derived in Eq. (22). (b) Absolute magnitude of the linear correction term $|b_1|$ vs. Δ . The empirical power-law fit ($\sim \Delta^{1.03}$) matches the analytical boundary scaling expectation of $\Delta^{1.00}$ (Eq. 23). (c) Deviation of the leading polynomial coefficient $|\hat{\theta}_0|$ as a function of Δ . The fitted scaling exponent ($\sim \Delta^{0.65}$) perfectly confirms the leading-order expansion $\hat{\theta}_0(\Delta) = \hat{\theta}_0^{(0)} + c_1 \Delta^{1-\nu} + O(\Delta)$ for $1 - \nu = 0.65$ (Eq. 25). Avoid titles, combine all of them in one figures with different colors + labels. Remove $\Delta < 10^{-7}$

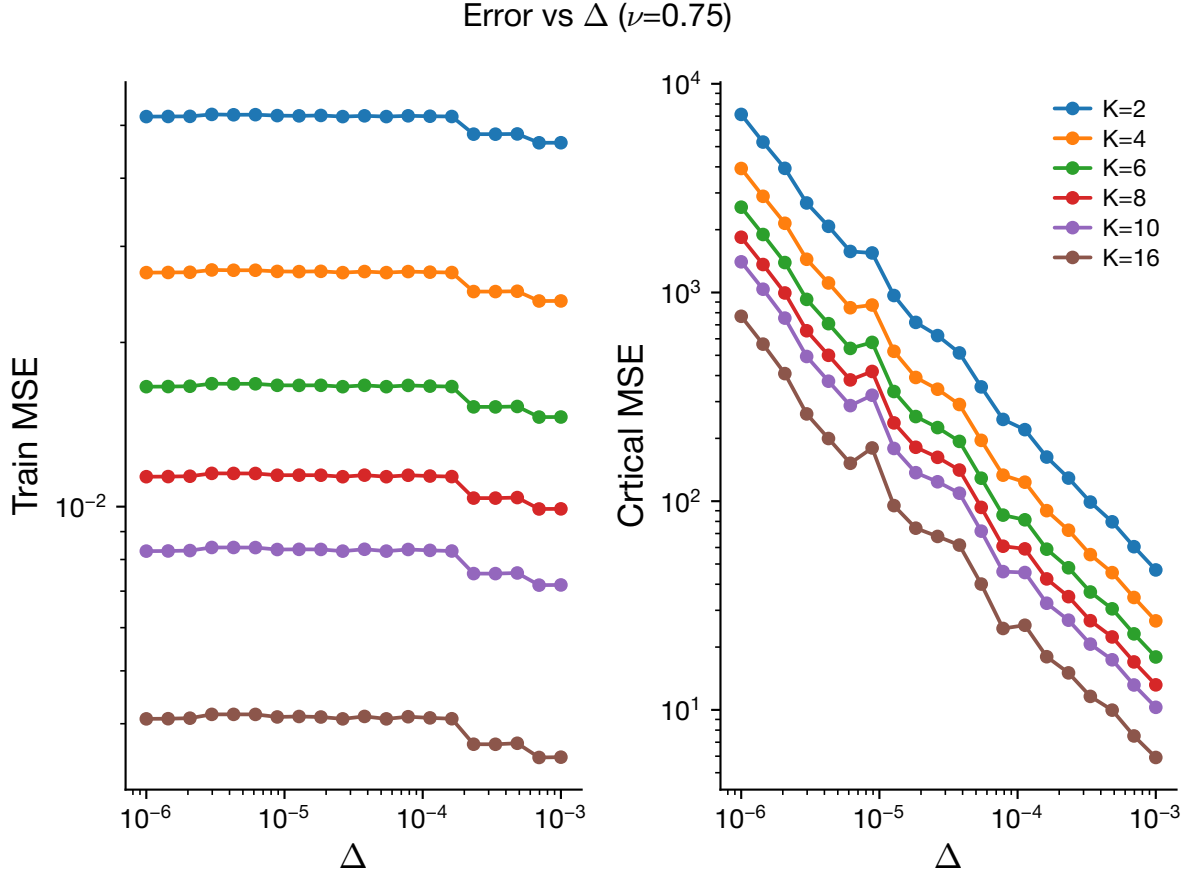


Figure 3: **Training and Critical Mean Squared Error (MSE) dependence on gap size Δ for $\nu = 0.75$.** **(Left)** Train MSE evaluated on the training distribution away from the gap. **(Right)** Critical MSE evaluated inside the excised gap near r_c across polynomial degrees $K \in \{2, 4, 6, 8, 10, 16\}$. Increasing model capacity K lowers overall approximation error, while the divergence rate as $\Delta \rightarrow 0$ remains governed by the critical exponent $1 - \nu$. **Combine this with previous figure. Reduce aspect ratio. Define mathematical variables - $\mathcal{L}$, $\mathcal{L}_{\text{mse}}$**

6 Finite sampling effects for a deterministic target

Now that we have arrived at the expected scaling for the infinite data sample limit, we move on to analyzing the scaling of θ in the case of finite data samples. We pose the problem as a ridge regression of the following form:

$$\mathcal{L} = \frac{1}{2} \sum_{\mu=1}^P [w(r_{\mu}) (\hat{\theta}^T \phi(r_{\mu}) - y_{\mu})^2] + \frac{\lambda}{2} \|\hat{\theta}\|^2 \quad (33)$$

What is the meaning of this loss-function, why is there a λ . Cite relevant references, why is it called the ridge loss. Do a toy-example to show how minimization works and put it as a boxes. The loss function in (17) has been modified to accommodate finite number of data points P.

Ridge Regression Explained

Core Idea: Standard linear regression tries to fit data points by minimizing prediction errors. However, when data is noisy, limited (P is small), or features are heavily correlated, standard regression often overfits—assigning massive, unstable weights $\hat{\theta}$ to match tiny noise fluctuations.

Ridge regression solves this by adding an L_2 **penalty term** ($\frac{\lambda}{2}\|\hat{\theta}\|^2$) that artificially penalizes large parameter magnitudes.

Physical Toy Example: Order Parameter Extraction near Criticality

Imagine estimating a soft mode $\hat{\theta}$ from d spatial sensors measuring local order $\phi(r_\mu)$. Near a critical point, neighboring sensors become strongly correlated due to diverging correlation lengths.

- **Ordinary Least Squares** ($\lambda = 0$): With finite time samples P , the empirical covariance matrix $\hat{\Sigma} = \frac{1}{P}\Phi^\top\Phi$ is ill-conditioned. Standard regression assigns unphysical, wildly oscillating weights to adjacent sensors ($\hat{\theta}_1 = +10^4, \hat{\theta}_2 = -10^4$) to fit sample noise.
- **Ridge Regression** ($\lambda > 0$): The penalty term prevents singular inversions $(\hat{\Sigma} + \lambda I)^{-1}$. It dampens high-frequency sample noise, smoothly spreading weights across the collective critical mode e_1 .

Key Takeaway for Finite Sampling Effects: In the finite data regime ($P < \infty$), λ acts as an effective noise floor that regularizes singular empirical modes, directly governing how fast $\hat{\theta}$ converges to the theoretical infinite-sample limit θ^* .

Differentiating the loss function with respect to $\hat{\theta}$ and setting to zero, we arrive at

$$\begin{aligned} \sum_{\mu=1}^P w(r_\mu) \phi(r_\mu) \phi(r_\mu)^T \hat{\theta} - \sum_{\mu=1}^P w(r_\mu) y_\mu \phi(r_\mu) + \lambda \hat{\theta} &= 0 \\ \Rightarrow \left(\frac{1}{P} \sum_{\mu=1}^P w(r_\mu) \phi(r_\mu) \phi(r_\mu)^T + \frac{\lambda}{P} I \right) \hat{\theta} &= \frac{1}{P} \sum_{\mu=1}^P w(r_\mu) y_\mu \phi(r_\mu) \end{aligned}$$

This motivates the definitions

$$\hat{M} = \frac{1}{P} \sum_{\mu=1}^P w(r_\mu) \phi(r_\mu) \phi(r_\mu)^T \quad (34)$$

and

$$\hat{b} = \frac{1}{P} \sum_{\mu=1}^P w(r_\mu) y_\mu \phi(r_\mu) \quad (35)$$

So the empirical estimator can be compactly written as

$$\hat{\theta}(\Delta, K, P) = (\hat{M} + \frac{\lambda}{P} I)^{-1} \hat{b} \quad (36)$$

$$\hat{b} \in \mathbb{R}^{K \times 1}; \quad \hat{M} \in \mathbb{R}^{K \times K}; \quad \hat{\theta} \in \mathbb{R}^{K \times 1} \quad (37)$$

Assume

$$y_\mu = g(r_\mu) \quad (38)$$

Therefore

$$\hat{b} = \frac{1}{P} \sum_{\mu=1}^P w(r_\mu) g(r_\mu) \phi(r_\mu) \quad (39)$$

and

$$\hat{M} = \frac{1}{P} \sum_{\mu=1}^P w(r_\mu) \phi(r_\mu) \phi(r_\mu)^T \quad (40)$$

Then the estimator is

$$\hat{\theta} = (\hat{M} + \frac{\lambda}{P} I)^{-1} \hat{b} \quad (41)$$

$$\Rightarrow \hat{\theta}(\Delta, K, P) = \left(\frac{1}{P} \sum_{\mu=1}^P w(r_\mu) \phi_\mu \phi_\mu^T + \frac{\lambda}{P} I \right)^{-1} \left(\frac{1}{P} \sum_{\mu=1}^P w(r_\mu) y_\mu \phi_\mu \right) \quad (42)$$

Define

$$M(\Delta, K) = \mathbb{E}_{r \sim \rho_\Delta}[\hat{M}] = \mathbb{E}_{r \sim \rho_\Delta}[w(r) \phi(r) \phi(r)^T] \quad (43)$$

$$b(\Delta, K) = \mathbb{E}_{r \sim \rho_\Delta}[\hat{b}] = \mathbb{E}_{r \sim \rho_\Delta}[w(r) g(r) \phi(r)] \quad (44)$$

As $P \rightarrow \infty$

$$\hat{\theta}(\Delta, K, P) \rightarrow \theta_\infty(\Delta, K) = M^{-1} b \quad (45)$$

which corresponds to the continuum solution obtained in the infinite-sample limit (see eqn (29)).

6.1 Perturbation around infinite sample limit

This has to be expanded. Give as much details as possible. Has been attempted below. Will modify according to feedback

We now introduce the finite data effect as a perturbation to the infinite limit case. That is, we introduce fluctuations in the quantities $M(\Delta, K)$ (eqn.27) and $b(\Delta, K)$ (eqn.28) as a function of the number of data samples P , around the infinite sample limit. Therefore the quantities mentioned above become:

$$\hat{M} = M(\Delta, K) + \delta M \quad (46)$$

and

$$\hat{b} = b(\Delta, K) + \delta b \quad (47)$$

where

$$\delta M = \frac{1}{P} \sum_{\mu=1}^P [w(r_\mu) \phi_\mu \phi_\mu^T - M(\Delta, K)]; \quad \delta b = \frac{1}{P} \sum_{\mu=1}^P [w(r_\mu) g(r_\mu) \phi_\mu - b(\Delta, K)] \quad (48)$$

Hence, in the finite sample limit, the quantity of interest is (see eqn.34)

$$\hat{\theta}(\Delta, K, P) = (M(\Delta, K) + \delta M + \frac{\lambda}{P} I)^{-1} (b(\Delta, K) + \delta b) \quad (49)$$

Since the problem involves a score-matching objective which is a quadratic loss function in terms of $\hat{\theta}$, to arrive at an explicit scaling of $\hat{\theta}$ with P , we compute the mean and variance of the key quantities in the following manner: **Why and how are you making these claims?** Has been addressed above

$$\begin{aligned} \mathbb{E}[\delta M] &= \mathbb{E}[\hat{M} - M] \\ &= \mathbb{E}[\hat{M}] - \mathbb{E}[M] \\ &= 0 \end{aligned} \quad (50)$$

and

$$\begin{aligned}
\mathbb{E}[\delta b] &= \mathbb{E}[\hat{b} - b] \\
&= \mathbb{E}[\hat{b}] - \mathbb{E}[b] \\
&= 0
\end{aligned} \tag{51}$$

From (46), we can define

$$\delta b = \frac{1}{P} \sum_{\mu=1}^P v_{\mu}, \quad \text{where } v_{\mu} \equiv w(r_{\mu})g(r_{\mu})\phi(r_{\mu}) - b \tag{52}$$

Each $v_{\mu} \in \mathbb{R}^K$ is an i.i.d zero-mean random vector as $\mathbb{E}_{r \sim \rho_{\Delta}}[w(r)g(r)\phi(r)] = b$. Now,

$$\begin{aligned}
\mathbb{E}[\delta b \delta b^{\top}] &= \mathbb{E} \left[\left(\frac{1}{P} \sum_{\mu=1}^P v_{\mu} \right) \left(\frac{1}{P} \sum_{\mu=1}^P v_{\mu}^{\top} \right) \right] \\
&= \frac{1}{P^2} \sum_{\mu=1}^P \sum_{\nu=1}^P \mathbb{E}[v_{\mu} v_{\nu}^{\top}]
\end{aligned} \tag{53}$$

$$\begin{aligned}
\implies \mathbb{E}[\delta b \delta b^{\top}] &= \frac{1}{P^2} \sum_{\mu=1}^P \mathbb{E}[v_{\mu} v_{\mu}^{\top}] \\
&= \frac{1}{P^2} \cdot P \cdot \mathbb{E}_{r \sim \rho_{\Delta}}[v(r) v(r)^{\top}] \\
&= \frac{1}{P} \text{Cov}_{r \sim \rho_{\Delta}}(w(r)g(r)\phi(r))
\end{aligned} \tag{54}$$

To evaluate the covariance matrix entrywise:

$$[\text{Cov}_{r \sim \rho_{\Delta}}]_{kl} = \int \rho_{\Delta}(r) (w(r)g(r)r^k - b_k)(w(r)g(r)r^l - b_l) dr \tag{55}$$

Since $w(r)g(r) = \rho_{\Delta}(r)\sigma_1(r) \cdot \frac{1}{\sigma_1(r)} = \rho_{\Delta}(r)$, this simplifies directly to the moments of the unweighted distribution:

$$[\text{Cov}_{r \sim \rho_{\Delta}}(w(r)g(r)\phi(r))]_{kl} = \int \rho_{\Delta}(r) r^{k+l} dr - b_k b_l = O(1) \tag{56}$$

Thus,

$$\mathbb{E}[\delta b \delta b^{\top}] = \frac{1}{P} \Sigma_b \sim O(P^{-1}) \tag{57}$$

Similarly we can proceed to see how the fluctuations in δM scale with P. From (46)

$$\delta M_{ij} = \frac{1}{P} \sum_{\mu=1}^P Z_{ij}(\mu), \quad \text{where } Z_{ij}(\mu) \equiv w(r_{\mu})\phi_i(r_{\mu})\phi_j(r_{\mu}) - M_{ij} \tag{58}$$

Each scalar $Z_{ij}(\mu)$ is zero-mean and i.i.d across samples μ . Taking the expected product of two matrix element perturbations δM_{ij} and δM_{kl} :

$$\mathbb{E}[\delta M_{ij} \delta M_{kl}] = \mathbb{E} \left[\left(\frac{1}{P} \sum_{\mu=1}^P Z_{ij}(\mu) \right) \left(\frac{1}{P} \sum_{\nu=1}^P Z_{kl}(\nu) \right) \right]$$

$$= \frac{1}{P^2} \sum_{\mu=1}^P \sum_{\nu=1}^P \mathbb{E}[Z_{ij}(\mu)Z_{kl}(\nu)] \quad (59)$$

Only the diagonal terms ($\mu = \nu$) contribute non-zero values. Therefore

$$\begin{aligned} \mathbb{E}[\delta M_{ij} \delta M_{kl}] &= \frac{1}{P^2} \sum_{\mu=1}^P \mathbb{E}[Z_{ij}(\mu)Z_{kl}(\mu)] \\ &= \frac{1}{P} \text{Cov}_{r \sim \rho_\Delta}(w(r)\phi_i(r)\phi_j(r), w(r)\phi_k(r)\phi_l(r)) \end{aligned} \quad (60)$$

Expanding the covariance integral for monomial basis $\phi_i(R) = r^i$:

$$\begin{aligned} \text{Cov}_{r \sim \rho_\Delta}(w(r)r^{i+j}, w(r)r^{k+l}) &= \int \rho_\Delta(r)(w(r)r^{i+j} - M_{ij})(w(r)r^{k+l} - M_{kl})dr \\ &= \int \rho_\Delta(r)w(r)^2 r^{i+j+k+l} dr - M_{ij}M_{kl} \end{aligned} \quad (61)$$

The convergence of the integral in $\text{Cov}(w(r)r^{i+j}, w(r)r^{k+l})$ depends explicitly on the value of the critical exponent ν . Because $w(r) = \rho_\Delta(r)\sigma_1(r) \sim |r|^{-\nu}$ near the gap edge, the integrand contains $w(r)^2 \sim |r|^{-2\nu}$. Therefore, the quantity $\mathbb{E}[\delta M \delta M^\top]$ scales as

$$\implies \text{Cov}_{r \sim \rho_\Delta}(w(r)\phi_i(r)\phi_j(r), w(r)\phi_k(r)\phi_l(r)) \sim \begin{cases} O(1), & \nu < 1/2 \\ \ln(1/\Delta), & \nu = 1/2 \\ \Delta^{-(2\nu-1)} & \nu > 1/2 \end{cases} \quad (62)$$

Having established the entrywise covariance of the empirical design matrix in Eq.60, we express δM as a stochastic matrix operator scaled by the Central Limit Theorem rate $P^{-1/2}$:

$$\delta M = \frac{1}{\sqrt{P}} \Xi(\Delta, \nu) \quad (63)$$

where $\Xi(\Delta, \nu) \in \mathbb{R}^{K \times K}$ is zero-mean Gaussian matrix random variable with $\mathbb{E}[\Xi] = 0$. The spectral norm of the perturbation scales strictly as:

$$\|\delta M\|_2 = O_p\left(\frac{1}{\sqrt{P}} \mathcal{F}(\Delta, \nu)\right) \quad (64)$$

with the gap-dependent fluctuation profile given explicitly by:

$$\mathcal{F}(\Delta, \nu) = \begin{cases} O(1), & \nu < 1/2 \\ O\left(\sqrt{\ln(1/\Delta)}\right), & \nu = 1/2 \\ O\left(\Delta^{-(\nu-1/2)}\right), & \nu > 1/2 \end{cases} \quad (65)$$

Show that fluctuations are of the $O(P^{-1/2})$. Has been addressed. $\mathcal{F}(\Delta, \nu)$ may need a little bit more detailed derivation

6.2 Trade-off Between Deterministic Gap Bias and Finite Data Statistical Noise

We can use the results from eqn.(55) and (63) to see the explicit scaling of the quantity in eqn.(47)

We define,

$$A = M + \frac{\lambda}{P} I \quad (66)$$

Then, applying Neumann series expansion to first order in $P^{-1/2}$,

$$\hat{\boldsymbol{\theta}}(\Delta, K, P) = (A + \delta M)^{-1}(b + \delta b) = (A^{-1} - A^{-1}\delta M A^{-1} + O(\delta M^2))(b + \delta b) \quad (67)$$

$$\implies \hat{\boldsymbol{\theta}} = A^{-1}b + A^{-1}\delta b - A^{-1}\delta M A^{-1}b + O(P^{-1}) \quad (68)$$

$$\implies \hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + A^{-1}\delta b - A^{-1}\delta M \boldsymbol{\theta} + O(P^{-1}) \quad (69)$$

$$\implies \hat{\boldsymbol{\theta}}(\Delta, K, P) = \boldsymbol{\theta}(\Delta, K) + \frac{1}{\sqrt{P}}\boldsymbol{\eta}(\Delta, K) + O(P^{-1}) \quad (70)$$

where

$$\boldsymbol{\eta}(\Delta, K) = \sqrt{P}A^{-1}(\delta b - \delta M \boldsymbol{\theta}) \quad (71)$$

The first and second moments of $\boldsymbol{\eta}$ evaluate to:

$$\mathbb{E}[\boldsymbol{\eta}] = \sqrt{P}\left(A^{-1}\mathbb{E}[\delta b] - A^{-1}\mathbb{E}[\delta M]\boldsymbol{\theta}\right) = 0 \quad (72)$$

$$\begin{aligned} \mathbb{E}[\boldsymbol{\eta}\boldsymbol{\eta}^\top] &= PA^{-1}\mathbb{E}\left[(\delta b - \delta M \boldsymbol{\theta})(\delta b - \delta M \boldsymbol{\theta})^\top\right](A^{-1})^\top \\ &= PA^{-1}\mathbb{E}\left[\delta b \delta b^\top - \delta b(\delta M \boldsymbol{\theta})^\top - (\delta M \boldsymbol{\theta})\delta b^\top + (\delta M \boldsymbol{\theta})(\delta M \boldsymbol{\theta})^\top\right]A^{-1} \end{aligned} \quad (73)$$

Since each expectation term in the bracket scales as $O(P^{-1})$, the factor of P cancels, leaving:

$$\mathbb{E}[\boldsymbol{\eta}\boldsymbol{\eta}^\top] = \mathcal{C}(\Delta, K) = O(1) \quad (74)$$

where $\mathcal{C}(\Delta, K)$ is the sampling covariance matrix of the estimator fluctuation.

Using these results, the first and second moments of the estimator $\hat{\boldsymbol{\theta}}(\Delta, K, P)$ are:

$$\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta}(\Delta, K) + O(P^{-1}) \quad (75)$$

$$\begin{aligned} \mathbb{E}[\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}^\top] &= \mathbb{E}\left[\left(\boldsymbol{\theta} + \frac{1}{\sqrt{P}}\boldsymbol{\eta} + O(P^{-1})\right)\left(\boldsymbol{\theta} + \frac{1}{\sqrt{P}}\boldsymbol{\eta} + O(P^{-1})\right)^\top\right] \\ &= \boldsymbol{\theta}\boldsymbol{\theta}^\top + \frac{1}{\sqrt{P}}\left(\mathbb{E}[\boldsymbol{\eta}]\boldsymbol{\theta}^\top + \boldsymbol{\theta}\mathbb{E}[\boldsymbol{\eta}^\top]\right) + \frac{1}{P}\mathbb{E}[\boldsymbol{\eta}\boldsymbol{\eta}^\top] + O(P^{-3/2}) \\ &= \boldsymbol{\theta}\boldsymbol{\theta}^\top + \frac{1}{P}\mathcal{C}(\Delta, K) + O(P^{-3/2}) \\ &= \boldsymbol{\theta}\boldsymbol{\theta}^\top + O(P^{-1}) \quad \text{to leading order in } P. \end{aligned} \quad (76)$$

Now, incorporating the scaling behavior with respect to the gap parameter Δ from Section 5, where $\boldsymbol{\theta}(\Delta) = \boldsymbol{\theta}_0 + \Delta^{1-\nu}\boldsymbol{\theta}_1 + O(\Delta^{3-\nu})$, we obtain the joint finite-sample and critical-gap scaling for the estimator:

$$\hat{\boldsymbol{\theta}}(\Delta, K, P) = \boldsymbol{\theta}_0 + \boldsymbol{\theta}_1\Delta^{1-\nu} + \frac{1}{\sqrt{P}}\boldsymbol{\eta}(\Delta, K) + O(\Delta^{3-\nu}, P^{-1}). \quad (77)$$

This explicitly separates the deterministic approximation error induced by the critical gap Δ from the $O(P^{-1/2})$ statistical fluctuations caused by finite dataset size P .

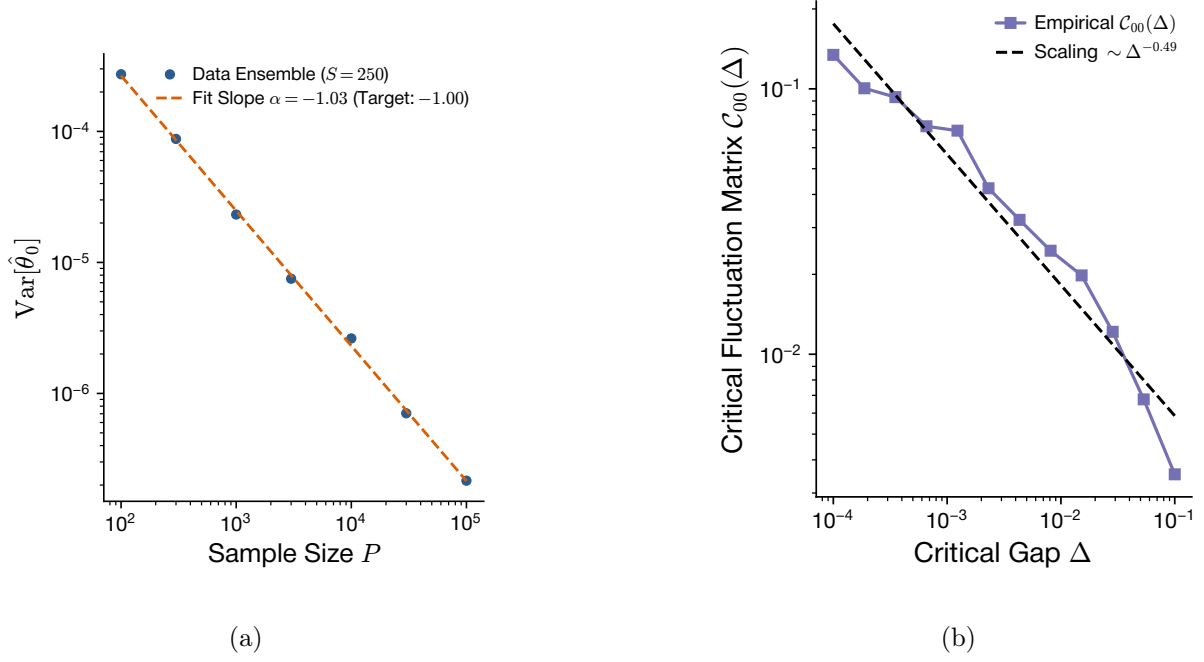


Figure 4: **Validation of Central Limit Theorem scaling and gap-dependent fluctuation amplification for finite sample size P .** (a) Variance of the estimated parameter $\text{Var}[\hat{\theta}_0]$ versus dataset size P over $S = 250$ independent data realizations. The empirical slope $\alpha = -1.03$ confirms the standard $O(P^{-1})$ CLT decay of estimator variance (Eq. 54). (b) Scaling of the empirical covariance matrix entry $\mathcal{C}_{00}(\Delta)$ with critical gap Δ . The power-law scaling $\mathcal{C}_{00}(\Delta) \sim \Delta^{-0.49}$ highlights the noise amplification driven by the ill-conditioning of the empirical design matrix $\hat{M}$ as $\Delta \rightarrow 0$.

7 Finite Sampling with noisy target

$$r_\mu \stackrel{i.i.d}{\sim} \rho_\Delta,$$

$$y = g(r_\mu) + \xi_\mu; \quad \xi \sim \mathcal{N}(0, \tau^2),$$

and

$$\phi(r) = (1, r, r^2, \dots, r^{K-1})^T$$